Neyman and Roughgarden (2022) propose a parameter value of $d = \sqrt{3}$. We adopt this convention in our work to avoid the risk of overfitting to the benchmarks we consider. Using Platt scaling on the arithmetic mean of k forecasts results in the lowest Brier score, as shown in Table 11.

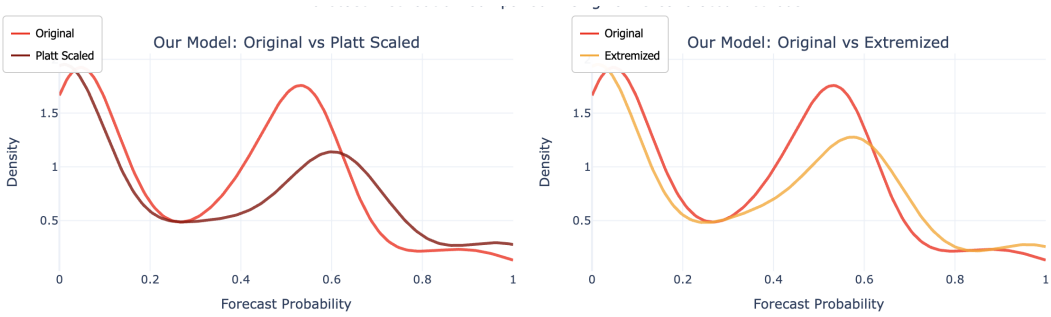

Figure 4: Scaling and extremization shifts the mass of probabilities toward the extremities, especially in the center mass which produces larger drops in Brier scores.

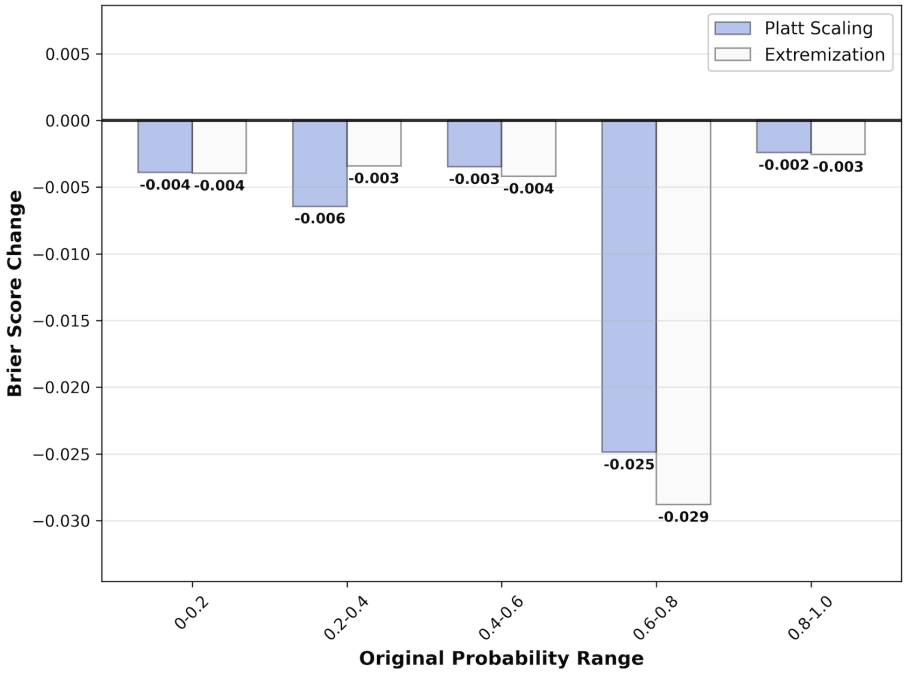

Figure 5: Across probability bins, the largest drops in Brier scores due to correction comes from the 0.6-0.8 forecast bin, followed by the 0.2-0.4 forecast bin.

Mechanisms of Improvement. As Table 11 shows, both Platt scaling and extremization significantly improve forecast accuracy, while isotonic regression and OLS are less competitive. Both Platt scaling and extremization effectively push probabilities toward their respective bounds (0 or 1). This suggests that initial forecast quality is critical to these kinds of statistical corrections, insofar as forecasts on the wrong side of 0.5 would be adjusted in the wrong direction. We also see that the improvement gained by original forecasts that are already confident and correct are (mechanically) limited. For example, an increase in the forecast from 0.995 to 0.999 yields a corresponding change in the Brier score of 2.4×10^{-5}. Meanwhile a change in the forecast from 0.497 to 0.381

yields a corresponding change in the Brier score of -0.102. That is, Platt scaling and extremization push forecasts towards the extremes. Figure 4 illustrates this redistribution effect, where mass shifts from the 0.5-0.6 range to the 0.6-0.75 range for events that ultimately resolve to 1.0.

This push toward the tails is what generates the substantial Brier score improvements when less confident but correct predictions become more confident, which we can also observe in Figure 5[17], where the largest Brier score decreases come from original forecasts $p \in [0.2, 0.4) \cup [0.6, 0.8)$.

Although extremization and Platt scaling are both effective and share mathematical similarities, supervisor agents offer significant improvements to forecast quality, as discussed in Section 7.2. The extremization approach detailed in Equation 2 requires aggregating k forecasts, which prevents us from leveraging the supervisor's contributions. Therefore, we adopt Platt scaling as our correction of choice: it preserves the advantages of extremization while enabling the supervisor resolution step, ultimately yielding more accurate forecasts as demonstrated in Table 11.

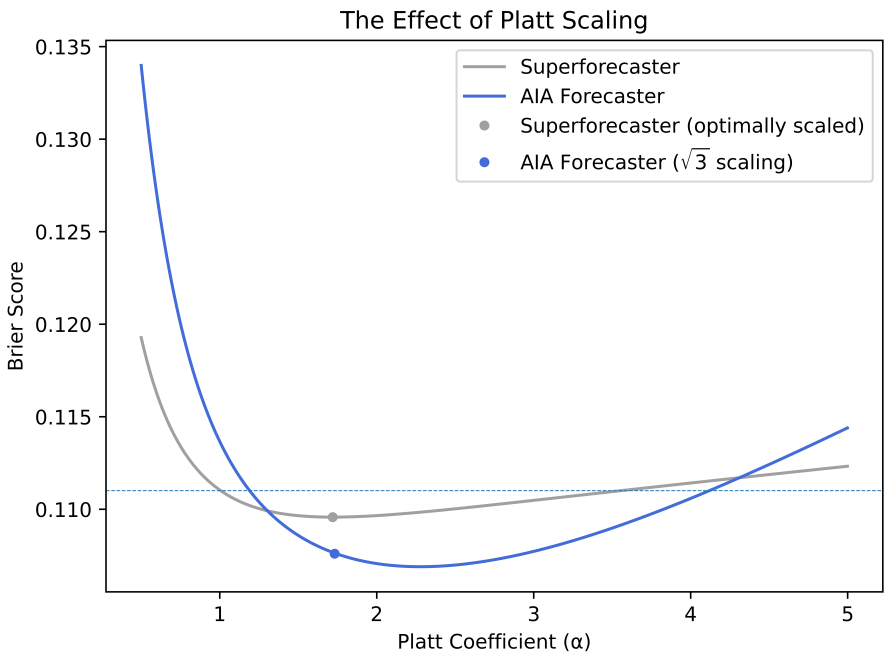

Figure 6: AIA Forecaster outperforms superforecasters across a range of Platt coefficients, including when the coefficient is optimized for superforecaster performance ($\alpha = 1.72$, gray dot). The AIA Forecaster uses a coefficient of $\alpha = \sqrt{3} \approx 1.73$, as suggested by Neyman and Roughgarden (2022) (blue dot).

Comparison with Superforecasters. Finally, because statistical corrections are agnostic to the technique used to generate the forecasts, they can also be applied to the superforecaster predictions. Indeed, applying a statistical correction to the superforecaster

17. Near $p \approx 0.5$, Platt scaling and extremization are approximately the identity function. In such scenarios, the brier score difference is minimally impacted as this effect dominates the quadratic Brier score improvement.

predictions is arguably required to produce a fair comparison between human and AI forecasts (Nikos et al., 2024).

In Figure 6, we plot the performance of both superforecasters and the AIA Forecaster across a range of Platt coefficients. As Figure 6 demonstrates, the AIA Forecaster outperforms the superforecaster baseline across a wide range of scaling parameters. Furthermore, scaling the AIA Forecaster with a Platt coefficient of $\sqrt{3} \approx 1.73$, as suggested by Neyman and Roughgarden (2022), outperforms superforecasters even when the scaling parameter is chosen to optimize their performance (a coefficient of 1.72). Optimizing the scaling coefficient for the AIA Forecaster yields a further modest improvement (at a coefficient of 2.27), though, as described above, we do not use this learned parameter to mitigate the risk of overfitting to a particular benchmark.

8 Discussion

This work describes a practical recipe for LLM-based judgmental forecasting: (i) prioritize evidence quality via agentic, adaptive search; (ii) reduce variance by ensembling independent runs; (iii) use a supervisor agent to actively reconcile disparate forecasts; and (iv) apply statistical calibration to counteract the behavioral biases of LLMs. Together, these components yield performance that is statistically indistinguishable from human superforecasters on ForecastBench, and diversifying to (albeit less accurate than) market consensus in liquid prediction markets.

Methodologically, we find that ensembling is not optional. Single-run performance is both noisier *and worse in expectation* than even simple averaging across a small number of independent generations. This also implies that the practice—which has thus far been common in the LLM forecasting literature—of neglecting to quantify uncertainty of LLM outputs themselves may threaten both internal and external validity of the claimed results. Second, generative verification methods are not necessarily superior to simple averaging. In fact, naive implementations of this approach can perform quite poorly. However, an agentic supervisor that can actively resolve disagreements and seek clarifications yields meaningful gains in forecasting performance. Third, statistical corrections provide a lightweight mechanism for aligning forecasts with scoring rules. However, post hoc calibration only helps when initial judgments are already on the correct side of 0.5, reinforcing that upstream evidence and reasoning are the primary mechanisms for improving forecast quality.

Limitations, risks and future work. A primary theme of this work is the difficulty of evaluating LLM-based forecasters. Perhaps the biggest challenge is the narrow temporal window in which this evaluation can be performed. On the one hand, events occurring before the model's knowledge cutoff are likely to have been memorized by the LLM (Sarkar and Vafa, 2024), making retrospective analysis difficult. On the other hand, evaluating the LLM on future events is inefficient—we need to wait for the events to occur!—which